THE EQUIVALENCE BETWEEN QUASI-COHERENT MODULES ON A SCHEME AND ON ITS SITES PRESERVES THE STANDARD PROPERTIES OF MODULES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement carries the author's own footnote listing the properties referred to by 'one of the properties of modules'; the footnote lies inside the counted statement span and is reproduced verbatim, and the card's statement depends on it.

Editorial note. The proof contains the author's parenthetical '(details omitted)' attached to the reduction of the coherent case to the finite type case; the reduction is stated with its reason and the parenthetical is reproduced verbatim, and the step is recorded as an obligation of the card.

Editorial note. The proof is a reduction scheme: the local properties are delegated to five named descent lemmas proved earlier in the chapter. That is the author's own organisation of the argument and it is retained as delivered; the division of the fifteen properties into global and local cases is the content the solver has to produce.

Editorial note. The counted spans carry 5 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The definitions of the global, local and flat properties of modules on a site. Modules on Sites, Definitions sites-modules-definition-global, sites-modules-definition-site-local and sites-modules-definition-flat, invoked in the statement at lines 2840-2843. Supplies the fifteen properties whose transfer is at issue.
- The equivalence between quasi-coherent modules on S and on the big and small sites given by the association F to F^a. Same chapter, Proposition proposition-equivalence-quasi-coherent, invoked in the statement at lines 2854-2855. Supplies the equivalences of categories whose compatibility with the properties is asserted.
- Quasi-coherence on a site is local for a covering. Modules on Sites, Lemma sites-modules-lemma-local-final-object, invoked at proof lines 2871-2872.

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Translates the assertion that the associated module has a local property into a condition on the members of a covering of the source.
- The descent lemmas for finite type, finite presentation, local generation by r sections, flatness and finite local freeness. Same chapter, Lemmas lemma-finite-type-descends, lemma-finite-presentation-descends, lemma-locally-generated-by-r-sections-descends, lemma-flat-descends and lemma-finite-locally-free-descends, invoked at proof lines 2874-2879. Discharge the local properties one by one once they have been translated into a statement on a covering.
- Coherent sheaves on a locally Noetherian scheme are the quasi-coherent sheaves of finite type. Cohomology of Schemes, Lemma coherent-lemma-coherent-Noetherian, invoked at proof lines 2880-2882. Reduces the exceptional case of coherence to the already handled finite type case.

Source reference. Original lemma 01XZ: sources/stacks-chow/tags.txt, source lines 2538–2538. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 05B2: sources/stacks-descent/pinned-original-source.tex, source lines 2222–2222. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 05B0: sources/stacks-algebra/pinned-original-source.tex, source lines 31681–31681. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 05AZ: sources/stacks-algebra/pinned-original-source.tex, source lines 31660–31660. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 05B1: sources/stacks-descent/pinned-original-source.tex, source lines 2208–2208. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 082U: sources/stacks-descent/pinned-original-source.tex, source lines 2185–2185. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original proposition 03DX: sources/stacks-descent/pinned-original-source.tex, source lines 2716–2716. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original definition 03ER: sources/stacks-chow/tags.txt, source lines 4466–4466. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original definition 03DE: sources/stacks-chow/tags.txt, source lines 4417–4417. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original definition 03DL: sources/stacks-chow/tags.txt, source lines 4424–4424. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 03DN: sources/stacks-chow/tags.txt, source lines 4426–4426. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme. Let $\tau \in \{Zariski, étale, smooth, syntomic, fppf\}$. Let $\mathcal{P}$ be one of the properties of modules*[1] *defined in Modules on Sites, Definitions 03DE, 03DL, and 03ER. The equivalences of categories*

$$QCoh(\mathcal{O}_S) \longrightarrow QCoh((Sch/S)_\tau, \mathcal{O}) \quad \text{and} \quad QCoh(\mathcal{O}_S) \longrightarrow QCoh(S_\tau, \mathcal{O})$$

defined by the rule $\mathcal{F} \mapsto \mathcal{F}^a$ seen in Proposition 03DX have the property

$$\mathcal{F} \text{ has } \mathcal{P} \Leftrightarrow \mathcal{F}^a \text{ has } \mathcal{P} \text{ as an } \mathcal{O}\text{-module}$$

except (possibly) when $\mathcal{P}$ is "locally free" or "coherent". If $\mathcal{P}$ ="coherent" the equivalence holds for $QCoh(\mathcal{O}_S) \to QCoh(S_\tau, \mathcal{O})$ when S is locally Noetherian and τ is Zariski or étale.

Proof. This is immediate for the global properties, i.e., those defined in Modules on Sites, Definition 03DE. For the local properties we can use Modules on Sites, Lemma 03DN to translate "$\mathcal{F}^a$ has $\mathcal{P}$" into a property on the members of a covering of X. Hence the result follows from Lemmas 05AZ, 05B0, 082U, 05B1, and 05B2. Being coherent for a quasi-coherent module is the same as being of finite type over a locally Noetherian scheme (see Cohomology of Schemes, Lemma 01XZ) hence this reduces to the case of finite type modules (details omitted). □

[1]The list is: free, finite free, generated by global sections, generated by r global sections, generated by finitely many global sections, having a global presentation, having a global finite presentation, locally free, finite locally free, locally generated by sections, locally generated by r sections, finite type, of finite presentation, coherent, or flat.